Homework 9 (Solver Practice)

March 31, 2026

This homework is a little bit shorter than we have previously seen. In this assignment, you will implement simplex method in its simplest form in Python. You will also use the solver to solve a problem. Both problems will be worth 30 points. **Please submit your code outputs to Crowdmark. In addition, please submit your code as a .ipynb file or screenshots of all code used to solve the problems.**

1)

a)

Implement the standard simplex method in Python (The one given in lecture 12). You are allowed to use the standard libraries, such as Numpy, but you cannot use the solver in your implementation. You code should be able to take in matrices/vectors $\vec{c}, \vec{b}, A$ corresponding to the standard form of the LP:

$$\text{minimize } \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} = \vec{b},\ \vec{x} \geq 0$$

Your code should output the optimal vector, $\vec{x}^*$, should it exist and it should output the value of $\vec{c} \cdot \vec{x}^*$ if $\vec{x}^*$ exists. You code will also need some method of initializing the simplex method and some method of pivoting rules (in order to avoid cycling).

b)

Use your code to solve the following LP

$$\text{minimize } \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} = \vec{b},\ \vec{x} \geq 0$$

where

$$A = \begin{bmatrix} 1 & 4 & 5 & 6 & 0 & 0 & 0 & 0 \\ 6 & 3 & 5 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & -1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

$$\vec{c} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{pmatrix}^T$$

and

$$\vec{b} = \begin{pmatrix} 8 & 7 & 6 & 5 & 4 & 3 \end{pmatrix}^T$$

c)

Check your solution to **b)** by plugging this LP into the solver. Be sure to include your code and output from your check.

2)

Suppose you have $\$100,000$ and you wish to maximize the amount of British Pounds (GPB) by some sequence of currency conversions. The exchange rates between currencies is given in the table below. Assume that the number in row i column j determines the amount of currency i that you will receive for 1 unit of currency j.

Exchange Rates	Dollars (USD)	Yen (JPY)	Euro (EUR)	Pounds (GBP)	Bitcoin (BTC)	Peso (MXN)
Dollars (USD)	1.0	0.00625	1.15224	1.33018	67150.9	0.0555
Yen (JPY)	159.915	1.0	184.269	212.723	10738400	8.8756
Euro (EUR)	0.86771	0.00543	1.0	1.1543	58267.4	0.04816
Pounds (GBP)	0.75164	0.0047	0.86613	1.0	50473.5	0.04172
Bitcoin (BTC)	0.00001	0	0.00002	0.00002	1.0	0
Peso (MXN)	18.0099	0.1126	20.7517	23.9563	1209380	1.0

(All exchange rates are from **https://www.oanda.com/currency-converter/en/?from=MXN&to=USD&amount=1** from March, 27th at approximately 5:15 PM). Also suppose there are limits to the amount of currency changes of a specific type that can occur. These limits are in the table below, with the number in row i column j referring to the maximum amount of currency that can be exchanged from currency j to currency i.

Exchange Rates	Dollars (USD)	Yen (JPY)	Euro (EUR)	Pounds (GBP)	Bitcoin (BTC)	Peso (MXN)
Dollars (USD)	0	1000000	10000	10000	1	10000
Yen (JPY)	10000	0	10000	10000	1	10000
Euro (EUR)	10000	1000000	0	10000	1	10000
Pounds (GBP)	20000	1000000	10000	0	1	10000
Bitcoin (BTC)	30000	0	10000	10000	0	0
Peso (MXN)	30000	1000000	10000	10000	1	0

a)

Model this problem as an LP (This does not need to be in standard form. General form is fine). Assume that fractions of currency can be traded and that there are no trading fees. Assume all trades occur instantly.

b)

Put your model into the solver. Report on the results. What is the maximum amount of GBP you can receive and what exchanges were needed in order to receive this amount of GBP?